

Assignment 7: Dialogue systems and question answering

NEXT UP: Submit assignment

Unlimited Attempts Allowed

(Read the whole module before you begin and manage your time between the tasks. Make reasonable assumptions as appropriate. Short and clear answers are preferred. Both members of the group must actively participate in every task. Each of you have up to 20 hours for a module including lectures.)

Dialogue systems and question answering

In this assignment, you will mainly consider **dialogue systems** and to some extent also **systems that can answer different kinds of questions**. Such systems have become very common in the form of digital assistants and chatbots. The lectures will partly be based on the following chapters from the book [Speech and Language Processing](#)

[Links to an external site.](#)

by Jurafsky and Martin:

- [Chatbots and dialogue systems](#)
- [Links to an external site.](#)
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- [Question answering](#)
- [Links to an external site.](#)
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In the introductory lecture we will mainly introduce dialogue systems, since this is what this week's mini-project is about.

Like in previous modules, this assignment may be solved in several different ways; the specific method chosen for the solution is secondary to the insights you gain in working with the assignment. It is therefore important that you describe the choices you made and your reasoning behind them.

Some of the tasks should be solved and reported as a group and some of them should be solved individually. **Please submit your work as a group.** In this assignment, **you are required to report in a pdf file and the code must be included in the same pdf file.**

1. Reading and summary [Group]

Read and briefly summarise the following paper:

- [GUS - a frame-driven dialog system](#)
- [Links to an external site.](#)
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This paper is an early work exhibiting several main ideas on which today's common digital assistants are based.

2. Implement a simple dialogue system [Group]

The purpose of this task is to make you aware of the basic steps and challenges involved in designing a digital assistant.

a) Implement a simple text-based digital assistant that can help with at least three things, for example:

- provide a weather forecast
- find a restaurant
- find the next tram/bus

However, implement in such a way that the assistant is extensible to a much larger set of tasks.

Note that it is possible to implement this in a trivial way by having the system fully control the flow like in "Enter 1 for account, 2 for insurance, 3 for loans", and continue in the same way with very specific questions. However, a dialogue system aims for a more flexible style, allowing for a more natural conversation. This is a non-trivial task, and while it is clearly not possible to fully solve in the scope of a weekly project, it is quite possible to get an understanding of many of the issues involved.

We suggest that you implement your assistant in the simplest possible way based on keyword matching (at least at first). Also think of the assistant as having access to back-end systems with the appropriate databases and functionality - for the purpose of this task you can simply use fictitious or random data.

As the output of this task, briefly describe how you have implemented the system, and especially any non-trivial functionality. Show some sample outputs, where the dialogue is as advanced as your system allows. Also show or discuss the limitations of your system. Submit your code in a zip-file separately from the module report.

b) Suggest how - if more time were available - you could make your dialogue system more advanced.

Self-check

- Have you answered all questions to the best of your ability?
- Is all the required information on the front page, is the file name correct etc.?
- Anything else you can easily check? (details, terminology, arguments, clearly stated answers etc.?)

Do not submit an incomplete module! We are available to help you, and you can receive a short extension if you contact us.